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## Jinmo Rhee

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## Education

- 2024 Ph.D. in Computational Design, School of Architecture, Carnegie Mellon University, US
- 2019 M.S. in Computational Design, School of Architecture, Carnegie Mellon University, US
- 2016 B.Arch. With Honors, College of Fine Arts, Korea National University of Arts, Korea

## Licenses

- 2016 Royal Institute of British Architects (RIBA) Part II, 66 Portland Palace, London W1B 1AD, England
- 2011 Royal Institute of British Architects (RIBA) Part I, 66 Portland Palace, London W1B 1AD, England

## Academic Appointments

- 2023 - Assistant Professor, School of Architecture Planning and Landscape, University of Calgary, Canada
- 2019 - 23 Graduate / Studio Instructor, School of Architecture, Carnegie Mellon University, US
- 2017 - 19 Teaching Assistant, School of Architecture, Carnegie Mellon University, US

## Experience

- 2023 - Director, [DESTECTIC \(Design Technology and Tactics\)](#), University of Calgary, Canada
- 2023 - Associate Director, [LID \(Laboratory for Integrated Design\)](#), University of Calgary, Canada
- 2020 - Co-Founders, [CRAIDL \(Creative AI and Design Launchpad\)](#), Carnegie Mellon University, US
- 2018 - 20 Manager, [CoDeLab \(Computational Design Laboratory\)](#), Carnegie Mellon University, US
- 2015 - 19 Interior and Branding Architect, JinmoRhee, Korea
- 2017.01 Project Architect Urban Operation, Korea
- 2013 - 17 Architectural Assistant, SSWA Architects, Korea
- 2015.07 Project Manager, Architectural Assistant, ArchiWorkshop, Korea
- 2013.09 Architectural Assistant, MLNP Architects, Korea
- 2013.04 Instructor of Rhinoceros3D and Grasshopper Workshop, Ein Architects & Engineering, Korea

## Funded Research

- 2026 - 28 PI with Prof. Alberto de Salvatierra at the Center for Civilization (CFC), *Return of Interest (ROI) Rural Area Modeling Project*, funded by Alberta Real Estate Foundation (AREF), CAD \$250,000
- 2026 - 27 PI, *An Interactive Ingress Prediction Tool for Instant Entry Flow Feedback in Early-Stage Layout Design*, funded by Mitacs and Perkins&Will, CAD \$30,000
- 2026 PI, *World-Model-Based Spatial Intelligence in Architectural Design Studio*, in-kind educational software support from World Labs for Marble access in studio teaching, estimated value USD \$2,250
- 2025 - 26 PI, *Research and Development of a Generative Layout System for Residential Floorplans*, funded by High-arc, CAD \$147,000
- 2025-2028 Co-PI with Prof. Alberto de Salvatierra at the Center for Civilization (CFC), *CCC x 801: An impact study of SAPL's presence in Downtown Calgary*, funded by City of Calgary and Calgary Downtown Association (CDA), CAD \$132,500
- 2025 - 27 Co-PI with Dr. Jason Hawkings, Dr. Alain Pietroniro, Dr. Lina Kattan, Prof. Alberto de Salvatierra, Dr. Seiran Heshami, Dr. Byron Andrew Miller *Social Innovation in Engineering for Climate-Neutral Cities*, funded by the Social Sciences and Humanities Research Council (SSHRC) through New Frontiers in Research Fund (NFRF), CAD \$250,000

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- 2025 – 26 PI, *Reimagining Generative Design Across Disciplines in the Era of AI*, funded by Transdisciplinary Scholarship Connector Grant from the University of Calgary, CAD \$8,500
- 2023 – 25 Co-PI with Prof. Alberto de Salvatierra at the Center for Civilization (CFC), *Return of Interest (ROI) Urban Modeling Pilot Project*, funded by Alberta Real Estate Foundation and the City of Airdrie, CAD \$260,000
- 2024 – 26 PI, *An Automated Camp Layout Technology for Rapid, Efficient, and Safer Wildfire Evacuee Settlement*, Amazon Web Services (AWS), Cloud Credit for Research, USD \$10,000
- 2023 – 24 PI, *An Automated Deployable Camp Layout Technology for Rapid, Efficient, and Safer Refugee Settlement Design: Proof of Concept Stage*, funded by VPR Catalyst Grants from the University of Calgary, CAD \$18,000
- 2020 – 23 PI, *Generative Design for Apartment Building Layout Using Artificial Intelligence Technology*, funded by DL E&C, USD \$220,000

## Refereed Publications

- 2026 **Jinmo Rhee**, Pedro Veloso, Bosco Chik, Charles Hawking, "Spatial Sculpting from Within: World Models and the Construction of Architecture through Connected Interior Spaces" in the *Association for Computer Aided Design in Architecture (ACADIA) Annual Conference 2026, Detroit* (in-review)
- 2026 Pedro Veloso, Katherine McLean, and **Jinmo Rhee**, "Design Thinking with Generative AI: Media Translation, Perception, and Immersion" in the *Association for Computer Aided Design in Architecture (ACADIA) Annual Conference 2026, Detroit* (in-review)
- 2026 **Jinmo Rhee**, Chapter 6. *Reconfiguring Design Agency in Artificial Intelligence: Language Interfacing Design Knowledge and Space* in *The Routledge Companion to Artificial Intelligence in Architecture*, Routledge (in-press)
- 2026 **Jinmo Rhee**, Chapter 5. Artificial Intelligence and the Changing Concept of the City: From Bounded Object to Spatial Aggregation, *Artificial Intelligence in Urban Planning and Design-Ed 2.*, Elsevier (in-press)
- 2026 Alberto de Salvatierra, Buyantegsh Naranbaatar, Giovanna Cross-Bussoli, Jasmine Keller, **Jinmo Rhee**, Kiana Rasti, Kiarash Kiany, Ruihong Chen "Rethinking Serious Games for Urban Systems: A Systematic Review Toward Iterative Citizen-Centered Gamification of Water and Transportation" in the *Sustainability* (in-review)
- 2026 Kiana Rasti, **Jinmo Rhee**, Spatial Simulation of Office-to-Residential Conversion: Building Form Type as a Spatial-Economic Criterion in Structuring Feasibility, *Journal of Building Engineering* (in-review)
- 2026 **Jinmo Rhee** and Ramesh krishnamurti "Language as Medium of Morphological Reasoning: Context-Aware Building Massing with LLMs" in the *Architecture, Special Issue: Computational Design and AI in Architecture: Methods, Evaluation, and Design Integration* (in-review)
- 2026 Manuel Ladron de Guevara, **Jinmo Rhee**, Ardavan Bidgoli, Vaidas Razgaitis, and Michael Bergin "Tokenizing Buildings: A Foundation Transformer For Layout Synthesis" in the *14th European Conference on Visual Information Processing(EUVIP), Luxembourg* (in-review)
- 2026 **Jinmo Rhee**, "From Tool to Design Companion: Asymmetric Human–Machine Collaboration in Architectural Design", *KAIST Research Series - Cultural DNA 2026*, Springer (in-press)
- 2026 **Jinmo Rhee** "Is Seoul Really a Republic of Apartment Complexes?: Urban Form Segregation and the Paradox of Diversity" in the *Environment and Planning B: Urban Analytics and City Science* (in-press)
- 2026 **Jinmo Rhee**, Kiarash Kiany and Alberto de Salvatierra "[Revealing Implicit Associations Between Urban Form and Socioeconomic Indices](#)" in the *Environment and Planning B: Urban Analytics and City Science*
- 2026 **Jinmo Rhee**, Eunjoo Oh, and Pedro Veloso "Computational Design as a Mode of Inquiry: Teaching Paradigms Unplugged" in the *Association of Collegiate Schools of Architecture (ACSA)'s 114th Annual Meeting, Chicago*
- 2026 Kiarash Kiany and **Jinmo Rhee**, "[Evaluating the Fiscal Performance of Urban Land Use Plans Through a Data-Driven Fitness Index](#)" in the *31th International Conference of the Association for Computer-Aided Architectural Design Research in Asia (CAADRIA), Taiwan*
- 2025 Pedro Veloso and **Jinmo Rhee**, "Shifting Agencies: Human–AI Co-Design of Dwelling Layouts via Deep Reinforcement Learning" in the *Association for Computer Aided Design in Architecture (ACADIA) Annual Conference 2025, Miami*
- 2025 **Jinmo Rhee**, "[Typological Knowledge-Driven Form Synthesis as a Design Method with Artificial Intelligence](#)" in the *Annual Conference of Architectural Institute of Korea, Seoul*. Awarded *Best Paper in the Computational Design Division*.

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- 2025 **Jinmo Rhee**, "Computational Design Education in Schools of Architecture: Exploring Curriculum Applicability through Global Cases" in the *Annual Conference of Architectural Institute of Korea*, Seoul
- 2025 **Jinmo Rhee**, Eunjo Oh, "Pedagogy Explorations into Alternative Use of Generative AI in Design Studios" in the *Technology | Architecture + Design (TAD) Journal*
- 2025 **Jinmo Rhee**, Pedro Veloso, "Efficient Camp Layout Design for Safer Wildfire Evacuee Solutions" in the *30th International Conference of the Association for Computer-Aided Architectural Design Research in Asia (CAADRIA)*, Tokyo
- 2025 **Jinmo Rhee**, "Comparative Analysis of Urban Structures in Three American Rust Belt Cities" in *Environment and Planning B: Urban Analytics and City Science*
- 2024 **Jinmo Rhee** and Ramesh Krishnamurti, "An Inductive Method for Classifying Building Form in a City with Implication for Orientation" in *Environment and Planning B: Urban Analytics and City Science*. Awarded the Michael Breheny Prize for the Best Paper
- 2023 **Jinmo Rhee** and Ramesh Krishnamurti, "Latent Space to Support Virtual 3D Models" in *Cultural Space on Metaverse: KAIST Research Series*
- 2023 **Jinmo Rhee** and Pedro Veloso, "Single Objective Optimization for Building Layout in an Apartment Complex" in *Annual Conference of Architectural Institute of Korea*, Jeongseon
- 2023 **Jinmo Rhee**, Pedro Veloso, and Ramesh Krishnamurti, "Three Decades of Machine Learning with Neural Networks in Computer-aided Architectural Design (1990-2021)" in *Design Science*
- 2023 Tomas Cabezon Pedroso, **Jinmo Rhee**, Daragh Byrne, "Feature space exploration as an alternative for design space exploration beyond the parametric space" in the *28th International Conference of the Association for Computer-Aided Architectural Design Research in Asia (CAADRIA)*, Navrangpura
- 2023 Michael Hasey, **Jinmo Rhee**, Daniel Cardoso Llach, "Form Data as a Resource in Architectural Analysis: An Architectural Distant Reading of Wooden Churches from the Carpathian Mountain Regions of Eastern Europe" in *Digital Creativity*
- 2023 **Jinmo Rhee** and Pedro Veloso, "A Physics-based Constraint Solver for the Site Layout Optimization of Non-convex Buildings with Multiple Setback Requirements" in the *20th International Conference of CAAD FUTURES*, Delft
- 2023 **Jinmo Rhee** and Bradley Williams, "Cost-Effective 3D Urban Massing Reconstruction of Public Aerial LiDAR Scans" in the *20th International Conference of CAAD FUTURES*, Delft
- 2022 **Jinmo Rhee**, Pedro Veloso, and Kim Jeongheon, "Graph-based Representation of Apartment Building Mass for Complex Layout" in *Annual Conference of Architectural Institute of Korea*, Jeju
- 2022 Dana Cupkova, Han Meng, and **Jinmo Rhee**, "Lithopic House: Computational Design Framework for re-Shaping Ecologically Attuned Architectural Systems out of Granular Matter" in the *42th Conference of Association for Computer Aided Design in Architecture (ACADIA)*, Philadelphia
- 2022 **Jinmo Rhee**, Pedro Veloso, Ardavan Bidgoli, Manuel Rodríguez, "Bubble2Floor: A pedagogical experience with deep learning for floor plan generation" in the *27th International Conference of the Association for Computer-Aided Architectural Design Research in Asia (CAADRIA)*, Australia
- 2022 **Jinmo Rhee**, Chapter 5. Urban Form Analysis through Morphometry and Machine Learning, *Artificial Intelligence in Urban Planning and Design-Ed 1.*, Elsevier
- 2021 **Jinmo Rhee** and Pedro Veloso, "Type Investigation in the Form of High-rise Building Using Deep Neural Network" in the *19th International Conference of CAAD FUTURES*, Los Angeles
- 2021 **Jinmo Rhee**, Pedro Veloso, Ramesh Krishnamurti, "Generative Design of Urban Fabrics Using Deep Learning" in the *26th International Conference of the Association for Computer-Aided Architectural Design Research in Asia (CAADRIA)*, Hongkong
- 2021 **Jinmo Rhee**, "Implementing Diffusion-Limited Aggregation Algorithm to Space Planning: A Design Experiment for Planning Underground Plaza Using Multi-agent System" in *Annual Conference of Architectural Institute of Korea*, Yeosoo
- 2020 **Jinmo Rhee**, Pedro Veloso, Ramesh Krishnamurti, "Integrating Building Footprint Prediction and Building Massing" in the *25th International Conference of the Association for Computer-Aided Architectural Design Research in Asia (CAADRIA)*, Bangkok
- 2019 **Jinmo Rhee** and Jae-won Chung, "A Study of Automation of Housing Design Method Using Artificial Intelligence" in *Annual Conference of Architectural Institute of Korea*, Daejeon, (Best Paper Awarded)
- 2019 **Jinmo Rhee** and Jae-won Chung, "Applicability of Artificial Intelligence in Apartment Complex Design" in *Annual Conference of Architectural Institute of Korea*, Daejeon
- 2019 **Jinmo Rhee**, Daniel Cardoso Llach, Ramesh Krishnamurti, "Context-rich Urban Analysis Using Machine Learning" in *Architecture in the Age of the 4th Industrial Revolution, eCAADe + SIGraDi*, Porto
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- 2019 Pedro Veloso, **Jinmo Rhee**, Ramesh Krishnamurti, "Multi-agent Space Planning A literature review (2008-2017)" in the *18th International Conference of CAAD FUTURES*, Daejeon
- 2016 **Jinmo Rhee**, "A study of Similar Quality of Transition Space between Korean Apartment House and Traditional House" in *Collections of Excellent Graduation Paper*, Architectural Institute of Korea
- 2011 Jiho Kim and **Jinmo Rhee**, "Computing Nature" in *Computing Aesthetics v.18-no.4*, Korea Information Processing Society

## Other Publications

- 2025 **Jinmo Rhee**, *Generative Design, How Does Artificial Intelligence Design?*, Ooribook
- 2024 Michael Hasey, **Jinmo Rhee** and Daniel Cardoso Llach, *Form data as a resource in architectural analysis: an architectural distant reading of wooden churches from the Carpathian Mountain regions of Eastern Europe* in *Critical Computational Relations in Design, Architecture and the Built Environment*, Routledge
- 2024 **Jinmo Rhee** and Ramesh Krishnamurti, *Latent Space to Support Virtual 3D Models* in *Cultural Space on Metaverse*, Springer Nature Singapore
- 2024 **Jinmo Rhee** and Pedro Veloso, *Digital Media Series: Python*, Amazon Kindle Publisher
- 2021 Pedro Veloso and **Jinmo Rhee**, "In Pursuit of Deep Architectural Design" in *Rumoer #76 Generative Design*, Building Technology Study Association
- 2020 **Jinmo Rhee**, *Context-rich Urban Analysis Using Machine Learning*, Eliva Press
- 2020 **Jinmo Rhee**, *Artificial Intelligence for Architecture*, Ooribook
- 2020 **Jinmo Rhee**, *Architectural Artificial Intelligence (Korean)*, Bookk
- 2020 **Jinmo Rhee** and Eddy Man Kim, *Digital Media Series: Grasshopper*, Amazon Kindle Publisher
- 2019 **Jinmo Rhee** and Eddy Man Kim, *Digital Media Series: Rhinoceros*, Amazon Kindle Publisher

## Exhibitions

- 2026 **Jinmo Rhee**, Curator and Organizer (with Emek Erdolu and Pedro Luís Alves Veloso), "*Reimagining Generative Design*," Exhibition + Symposium, Richardson Memorial Hall Gallery, Tulane University School of Architecture, New Orleans, LA, USA, April 3–20, 2026
- 2021 Exhibition Assistant, "*Vers un imaginaire numérique*," Montréal's Design Centre, Montréal, Canada
- 2021 **Jinmo Rhee**, "*Rethinking Urban Form: Morphological Understanding of City Space through Data*," Video work, Austria Pavilion, Venice Biennale
- 2019 **Jinmo Rhee**, Invited Artist, "*Context-rich Urban Analysis and Generation Using Artificial Intelligence*," The 2nd Special Exhibition for Korean Students in America, Korean Institute of Architects, Culture Station Seoul 284, Seoul, Korea
- 2015 "*Skydium and Jamsil Sports Complex Renovation Master Plan*," Korea National University of Arts, Seoul, Korea
- 2013 **Jinmo Rhee**, Invited Artist, "*Aesthetics of the Nature*," Munji Culture Center, Seoul, Korea
- 2013 Entry Team Artist, "*REAL DMZ 2013*," Art Sonje Center, Seoul, Korea

## Courses

Assistant Professor at University of Calgary

- Winter 2026 Introduction to Computational Design (DSGN 263), School of Architecture Planning and Landscape, University of Calgary, Canada
- Winter 2025
- Winter 2026 Computational Design for Built Environment (DSGN 363), School of Architecture Planning and Landscape, University of Calgary, Canada
- Winter 2025
- Winter 2026 Senior Research Studio in Architecture II (ARCH 702), School of Architecture Planning and Landscape, University of Calgary, Canada
- Winter 2024
- Winter 2026 Studio Research Selective (ARCH 690), School of Architecture Planning and Landscape, University of Calgary, Canada
- Fall 2025 Scripting and Generative Design (ARCH 680.29), School of Architecture Planning and Landscape, University of Calgary, Canada
- Fall 2024
- Fall 2023

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- Fall 2024 Intermediate Architectural Design Studio (ARCH 606), School of Architecture Planning and Landscape, University of Calgary, Canada
  - Winter 2024 Data Processing for Computational Design (ARCH 680.30), School of Architecture Planning and Landscape, University of Calgary, Canada
  - Winter 2024 Design with Deep Learning (ARCH 672.32), School of Architecture Planning and Landscape, University of Calgary, Canada

#### Graduate Instructor at Carnegie Mellon University

- Spring 2023 Generative Systems for Design (68-702), School of Architecture, Carnegie Mellon University, US
- Spring 2022
- Fall 2022 Scripting and Parametric Design (48-724), School of Architecture, Carnegie Mellon University, US
- Fall 2021
- Fall 2020
- Spring 2022 Introduction to Artificial Intelligence in Design Studies (48-770), School of Architecture, Carnegie Mellon University, US
- Fall 2021
- Spring 2021
- Spring 2021 Shaping Form (62-212), Carnegie Mellon University Qatar Campus, Qatar
- Spring 2022 Descriptive Geometry (62-175), School of Architecture, Carnegie Mellon University, US
- Spring 2021
- Spring 2021 Fundamentals of Computational Design (62-275), School of Architecture, Carnegie Mellon University, US
- Spring 2020
- Fall 2020 Architecture Design Studio: Foundation I (48-100), School of Architecture, Carnegie Mellon University, US
- Fall 2019

#### Teaching Assistant at Carnegie Mellon University

- Fall 2019 MSCD Pre-Thesis II (48-716), School of Architecture, Carnegie Mellon University, US
- Fall 2019 Digital Skill Workshop 2018 and 2019 (48-689), Carnegie Mellon University, US
- Fall 2018
- Fall 2019 Digital Media I and II (48-120, 48-125, 62-122, 62-123), Carnegie Mellon University, US
- Fall 2018
- Fall 2017 Generative Modeling (48-783), Carnegie Mellon University, US

### Invited Lectures

- 2026 Keynote Speaker at Symposium, "Shifting Design Agency at the Confluence of Learning Paradigms and Generative Design", Tulane University School of Architecture, New Orleans, LA, USA,
- 2025 Guest Speaker, "Generative Design: How Does Artificial Intelligence Design?", Department of Architecture & Architectural Engineering, Seoul National University, Korea
- 2025 Guest Speaker, "Witnessing Culture Technology in an Era of Collapsing Boundaries: Where Design, Space, Technology, and the Socio-Cultural Intersect", Graduate School of Culture Technology, KAIST, Korea
- 2025 Guest Lecturer for LAND 201, "Computational Urban Morphology: How Are Urban Spaces Read and Planned Through Data?", School of Architecture Planning and Landscape, University of Calgary, Canada
- 2024 Discussion Panel, "AI, Friend or Foe?", d.talks, Calgary, Canada
- 2024 Discussion Panel, "Hop on the Train or Get Left Behind: The Inevitable Amalgamation of Architects' Ingenuity & Artificial Intelligence?", BUILDDEX Alberta, Calgary, Canada
- 2024 Guest Lecturer for DSGN 231, "Computational Urban Morphology: How Are Urban Spaces Read and Planned Through Data?", School of Architecture Planning and Landscape, University of Calgary, Canada
- 2024 Guest Lecturer, "Learning-based System in Design, Beyond Oracle", Graduate School of Culture Technology, KAIST, Korea
- 2023 Guest Lecturer for 62-706 Generative Systems for Design, "Learning-based System in Design, Beyond Oracle", School of Architecture, Carnegie Mellon University, US
- 2023 Guest Lecturer for ARCH 8070 RAS Theory, "Learning-based System in Design, Beyond Oracle", Stuart Weitzman School of Architecture, University of Pennsylvania, US
- 2021, 2022 Guest Speaker for Graduate Seminar: Situating Research, "Analysis of Urban Form Using Artificial Intelligence", School of Architecture, Carnegie Mellon University, US

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- 2021 Discussion Panel for Exchange: AI in Design, School of Architecture, Carnegie Mellon University, US
  - 2017 Guest Lecturer for Intermediate Design Studio, "Digital to Physical", Sookmyung Women's University, Korea
  - 2017 Guest Speaker for Advanced Design Studio, "Digital to Physical", Korea National University, Korea

## Student Advising and Award

- 2026 - 30 PhD Student Supervisor in Environmental Design (Computational Design Focus), Michael de Haan, University of Calgary, Canada
- 2026 - 30 PhD Student Supervisor in Environmental Design (Computational Design Focus), Kiarash Kiany, University of Calgary, Canada
- 2026 Supervisor, Natural Sciences and Engineering Research Council of Canada (NSERC) Undergraduate Student Research Award (USRA) recipient Sean Luard, "The City at the Table: Consumer Behaviour Mapped Through Simulation", University of Calgary, Canada
- 2025 Finalist, Baglietto Furniture Design Competition—student project (Kiarash Kiany and Kiana Rasti)
- 2025 PhD Defense Neutral Chair, "Gamified Design Configurator: A Performance-Aware, Data-Driven Framework for Enabling Informed Decision-Making by Non-Experts in Early-Stage Design", Arman Khalilbeigi Khameneh, University of Calgary, Canada
- 2024 - 26 Master Student Supervisor in Design Research (Computational Design Focus), Kiana Rasti, University of Calgary, Canada
- 2024 - 25 Master Student Supervisor in Design Research (Computational Design Focus), Kiarash Kiany, University of Calgary, Canada
- 2024 - 25 PhD Supervisory Committee Member, Alex Sidaway, University of Calgary, Canada
- 2024 DDes Defense Neutral Chair, "Entangled Methodologies for Participatory Multispecies Futures", Priyanka Bista, University of Calgary, Canada
- 2024 PhD Defense Examiner, "Multi-objective Optimization of Building Envelopes for Embodied and Operational Greenhouse Gas Emissions", Negar Badri, University of Calgary, Canada
- 2024 Master Thesis Committee Member, "Design with data-driven shape rule recognition through learning-based shape grammar interpreter", Yujeong Seo, Carnegie Mellon University, US
- 2024 Master Thesis Committee Member, "Speculative City through the Ideation of Spaces in Metaverse", Kenneth Anggara, University of British Columbia, Canada
- 2023 Research Advisor, "Identifying Building Types Using Deep Learning", Zhiqi Pang, University of Calgary, Canada
- 2023 Master Thesis Committee Member, "Decrypting the Black Box: a designer's journey into the infinite possibilities of ML in design", Tomas Cabezon Pedroso, Carnegie Mellon University, US
- 2023 Master Thesis Committee Member, "Building Information Twinning: Investigating the Software Architecture of a Building's Digital Twin", Brad Williams, Carnegie Mellon University, US
- 2022 Master Thesis Committee Member, "The 3D Form Analysis of Regional Architecture Using Deep Learning: A Case Study of Wooden Churches from the Carpathian Mountains", Michael Hasey, Carnegie Mellon University, US

## Awards and Recognition

- 2025 The Michael Breheny Prize for the Best Paper, *Environment and Planning B: Urban Analytics and City Science*
- 2025 Best Paper in Annual Conference of Architectural Institute of Korea
- 2021 Grand Prize in Digital Architecture Competition 2021 (DA 2021), Architectural Institute of Korea and Korea Ministry of Land, Infrastructure and Transport
- 2021 - 22 Fellowship, Architectural Design Human Resource Development Project, Korea Agency for Infrastructure Technology Advancement
- 2020 Young CAADRIA Award 2020, The Association for Computer-Aided Architectural Design Research in Asia
- 2019 Grand Prize in Digital Architecture & Architecture Startup Competition 2019 (DAAS 2019), Architectural Institute of Korea and Korea Ministry of Land, Infrastructure and Transport
- 2019 Best Paper in Annual Conference of Architectural Institute of Korea, 2019
- 2019 SoA + GSA International Conference Travel Grants, Carnegie Mellon University

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- 2018 - 22 Graduate Student Small Project Help (GuSH) funds, Carnegie Mellon University
  - 2017 - 19 Merit-based Admission Scholarship, Carnegie Mellon University
  - 2016 Honorable Mentions in Collections of Excellent Graduation Paper, Architectural Institute of Korea
  - 2016 Valedictorian, Korea National University of Arts
  - 2015 Kookmin Bank Academic Scholarship
  - 2014 Honorable Mentions in Architectural Structure Design Competition, The Korean Structural Engineers Association
  - 2010 - 15 Academic Performance Scholarship, Korea National University
  - 2010 Honorable Mentions in Jeju Architecture Awards, Jeju City Government
  - 2010 Honorable Mentions in Idea Competition of the Public Facility on Kwanghwamoon Plaza, Seoul Design Foundation

## Academic Services

- 2026 - 2027 Organizing Chair, *CAAD Futures 2027 Conference*, University of Calgary
- 2026 Scholarship Adjudication Committee Member, *Faculty of Graduate Studies (FGS)*, University of Calgary
- 2026 Hiring Search Committee Member (Computation and Making), *M.Arch Program, School of Architecture, Planning and Landscape (SAPL)*, University of Calgary
- 2026 Hiring Search Committee Member (Building Science), *M.Arch Program, School of Architecture, Planning and Landscape (SAPL)*, University of Calgary
- 2026 Hiring Search Committee Member (History and Theory), *M.Arch Program, School of Architecture, Planning and Landscape (SAPL)*, University of Calgary
- 2026 - Reviewer, *Canada Foundation for Innovation*
- 2026 - Reviewer, *Journal of Urban Affairs*
- 2025 - Reviewer, *Automation in Construction*
- 2025 - Reviewer, *Automation in Construction*
- 2025 - Reviewer, *Association of Collegiate Schools of Architecture(ACSA) Annual Conference*
- 2025 - Reviewer, *Discover Sustainability*
- 2024 - Reviewer, *Journal of Mechanical Design*, The American Society of Mechanical Engineers
- 2024 - Reviewer, *International Journal of Architectural Computing*, CAAD-Futures Foundation
- 2024 - Reviewer, The Association for Computer Aided Design in Architecture (ACADIA) Conference
- 2024 Scientific Chair, the Association for Computer Aided Design in Architecture (ACADIA) Conference 2024, University of Calgary
- 2022 - Reviewer, *Environment and Planning B: Urban Analytics and City Science*, Design Society
- 2021 Session Chair in Urban Analytics, the 26th International Conference of the Association for Computer-Aided Architectural Design Research in Asia (CAADRIA), Hongkong
- 2019 - Reviewer, The Association for Computer-Aided Architectural Design Research in Asia (CAADRIA) Conference
- 2019 - Reviewer, The Ibero-American Society of Digital Graphics (SIGRADI)

## Certificates

- 2018 Mathematics for Machine Learning: Linear Algebra, Certificated by Imperial College London and offered through Coursera
- 2018 Mathematics for Machine Learning: Multivariate Calculus, Certificated by Imperial College London and offered through Coursera

## Patents

- 2024 Media Control Device and System, (*Patent in USA, 11910130*)
- 2023 Building Layout Automation Reinforced by Physics Simulation as Constraints Solver, (*In the process of provisional patent in USA / Canada / Republic of Korea*)

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- 2018 Diagrammatic Image Dataset Generation with Neural Networks, (*Provisional Patent in USA, 9470103699300040518560*)
- 2018 Custom Generative Systems with Adaptation Mechanisms, (*Provisional Patent in USA, 9470103699300040518553*)

## **Memberships**

- 2022 - Faculty Member, Association of Collegiate Schools of Architecture (ASCA)
- 2016 - Associate Member, Royal Institute of British Architects (RIBA)
- 2016 - Chartered Member, Architectural Institute of Korea (AIK)
- 2019 - Chartered Member, Society of Computational Design and Engineering Korea